

THE DEBUGGERS · SOFTWARE ENGINEERING

SecondHelping

PROGRESS REPORT

Campus Surplus-Food Marketplace · October 20–27, 2026

Zamo Rzgar Ahmed · Geming (Kevin) Wu · Bianca Zavadisk · Jadlaoui Marwen · Gabriel Fer Silva

WEEK · OVERVIEW

Three features shipped, one spec ready for review.

An AI chat assistant, a notification system with timed pickup reminders, and AI meal photo matching — plus a full documentation overhaul for peer review.

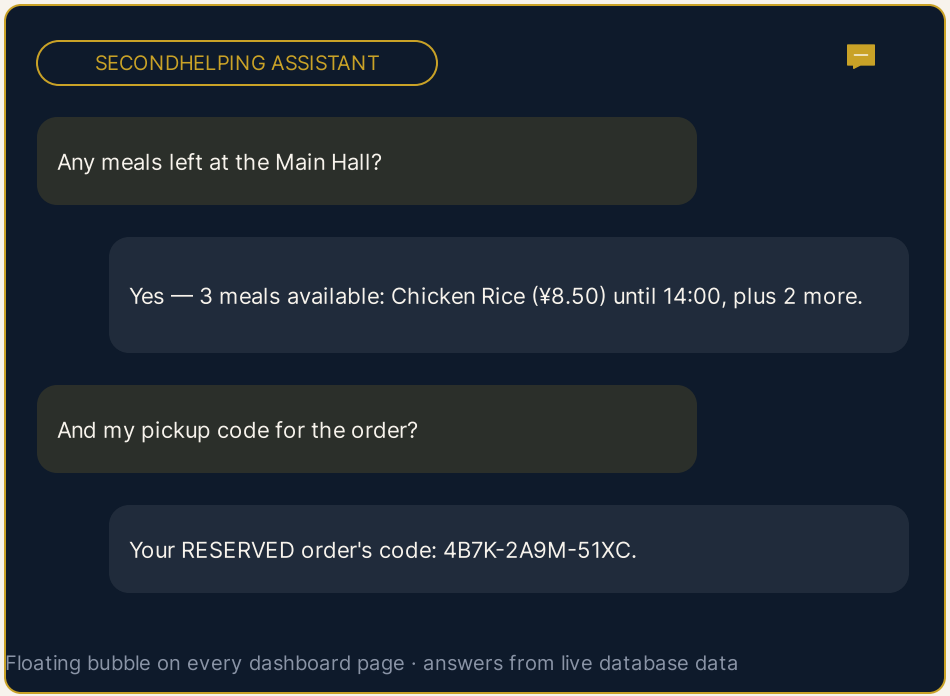
FIGURE 1 · WEEK 6 MILESTONE TIMELINE — OCT 20 → OCT 27



Three features + SRS v1.0 — the week moved from prototype to product.

Source: GitHub · bhsei/26C, git log, Oct 20–27 2026

Ask the marketplace — answers come from live data.



WHAT IT ANSWERS

· Available meals and dining halls
· Own orders and pickup codes
· Allergens and the current time

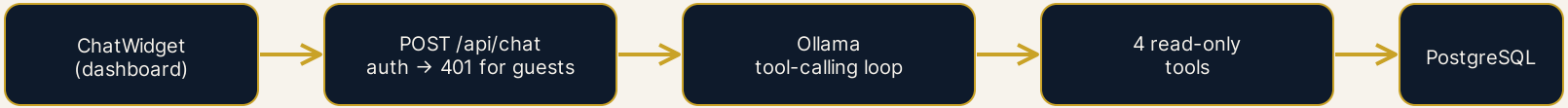
HOW IT BEHAVES

· Small talk and general questions, no tools
· Facts from the database, not model memory
· 100% local via Ollama — no API keys

Source: app/api/chat · lib/ollama.ts · lib/chat-tools.ts

A read-only tool-calling loop over live data.

FIGURE 2 · CHAT ASSISTANT ARCHITECTURE



Tools: get_available_listings · list_dining_halls · get_my_orders · get_current_time — all read-only

TABLE 1 · MODEL EVALUATION — TESTED LIVE

Model	Uses live data	Handles "thanks"	Verdict
llama3.2 (3B)	Mostly	Repeats meal list	Too small
mistral:7b	Ignores tools, invents meals	Repeats list	Rejected
qwen2:7b	**Reliable, real DB data**	**Correct short reply**	**Selected**

Pickup codes exposed only for RESERVED orders.

The assistant is **read-only** — it never places or pays for orders.

Default model: qwen2 : 7b

Source: app/api/chat · lib/chat-tools.ts · lib/ollama.ts

Every order event lands in the bell.

THE NOTIFICATION PANEL



Unread-count badge · newest-first panel · mark-as-read

TABLE 2 · EVENT TRIGGERS

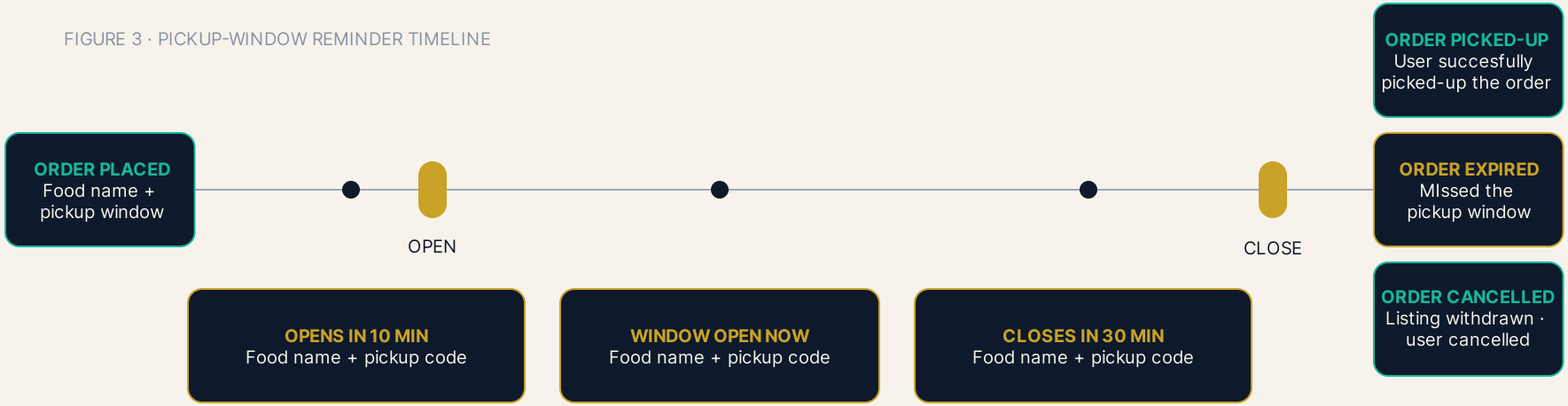
Event	Student	Supervisors
Reservation placed	Yes	Yes • listing creator
Pickup confirmed (with time)	Yes	—
Missed pickup window (expired)	Yes	Yes • whole hall
Listing withdrawn (cancelled + refunded)	Yes	Yes • whole hall

Hall-wide alerts go to all supervisors.

Source: app/api/notifications · lib/notifications.ts · lib/order-lifecycle.ts

Three timed reminders bracket every pickup window.

FIGURE 3 · PICKUP-WINDOW REMINDER TIMELINE



Scheduler sidecar (cron.js) pings the sweep endpoint every 60 s. Runs automatically with `pnpm dev`.

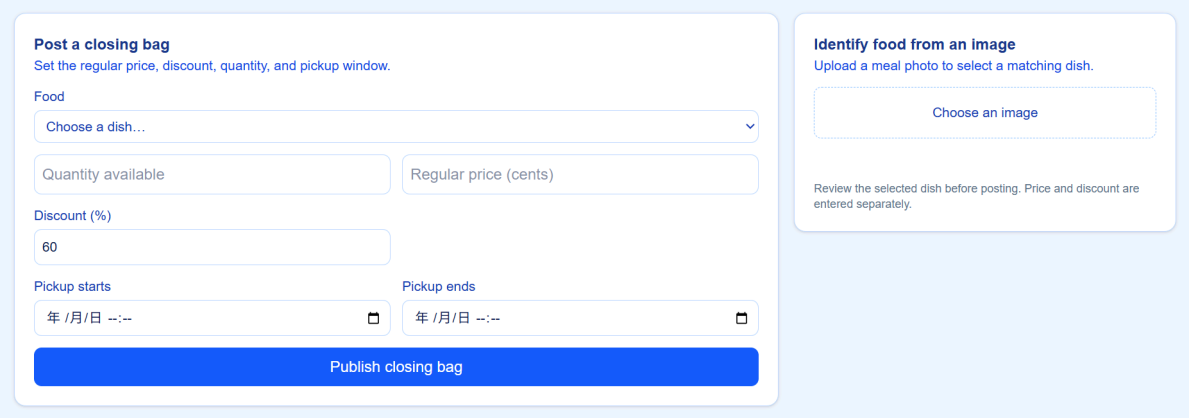
Per-order flags — reminderOpenSent · reminderCloseSent
Each reminder is sent exactly once.

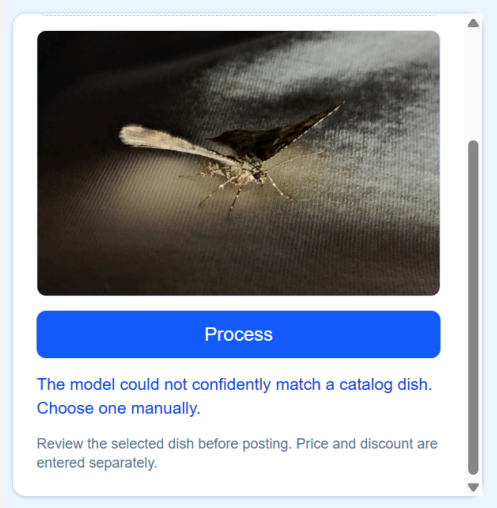
Engineering note: cron.js crash fixed Oct 27 — a statement misplaced inside the fetch chain (ASI bug)

Source: app/api/notifications/route.ts · cron.js

A photo becomes a catalog selection.

- Having many meals with similar names in the food list makes staff confused
- We used **qwen2.5vl:3b** with 3 billion parameters
- Prompt engineering to force the model to generate either a **food ID** or a **NULL**





Source: app/api/food-image-match · app/scripts/classify_meal.py · docs/SRS.md REQ-FUNC-011

Regarding the details.

Only supervisors or administrators can access this feature, since students cannot publish new listings.

Every time the model outputs a food ID, we ensure if the ID really exists, otherwise, a warning message is shown.

The JSON file containing all food IDs for the model to check is automatically created as a temporal file when the function is called. When the processing finishes, the temporal file is deleted.

First working slice of the SRS **Multimodal Listing Assistant (REQ-FUNC-011)** — now **partially implemented**

Source: app/api/food-image-match/route.ts · docs/sample_food_images/

A specification ready for peer review.

0.3 ➛ 1.0

SRS VERSION THIS WEEK

Chat assistant, image matching, notifications, and the payment lifecycle — all documented and verified against the running app.

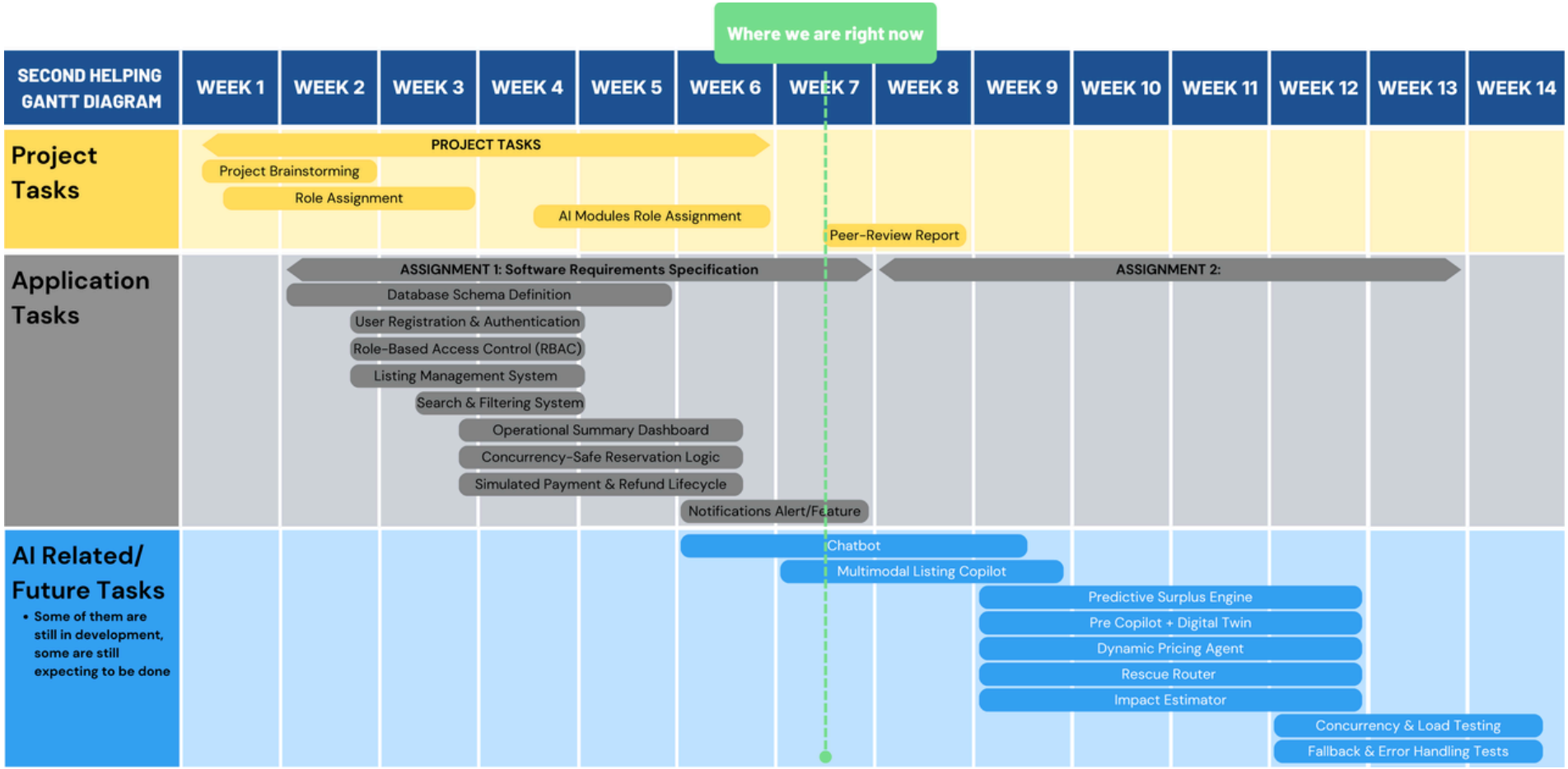
17 functional requirements
Each with status, MoSCoW priority, preconditions, and numbered acceptance criteria — statuses verified against the real app.

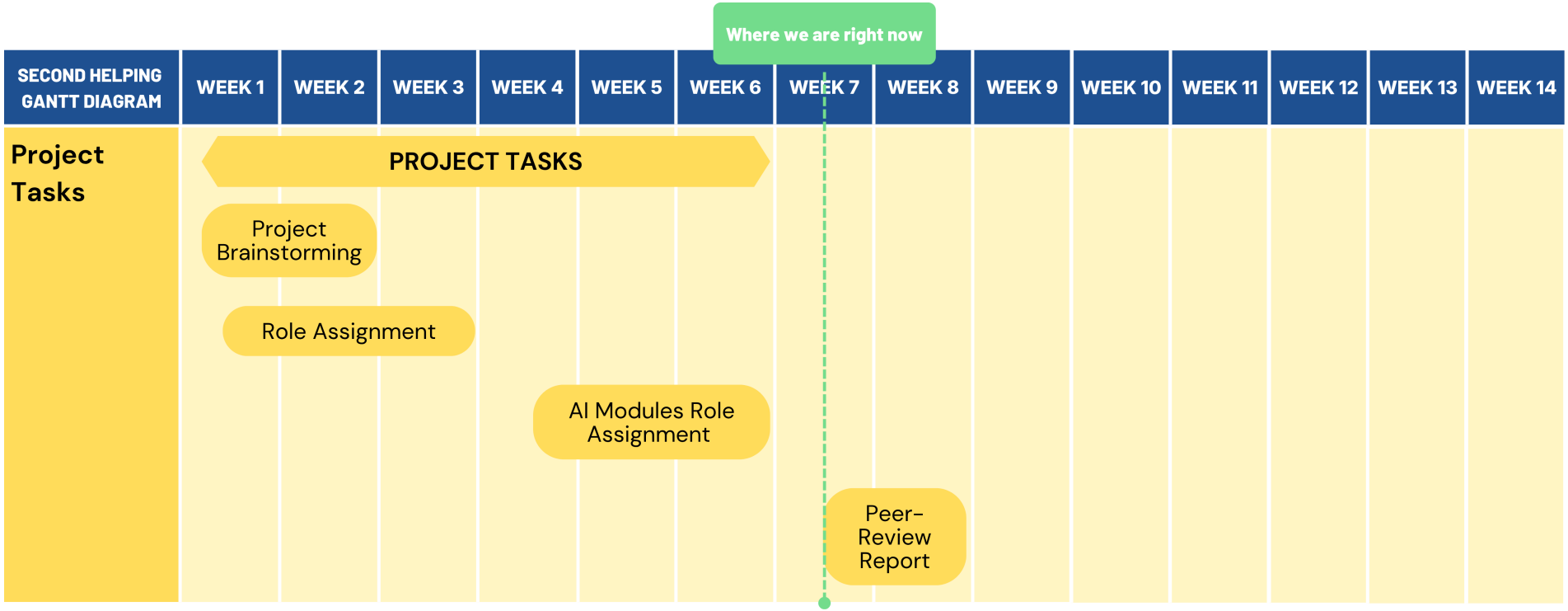
Verification traceability table
Every implemented requirement mapped to a test activity and expected evidence.

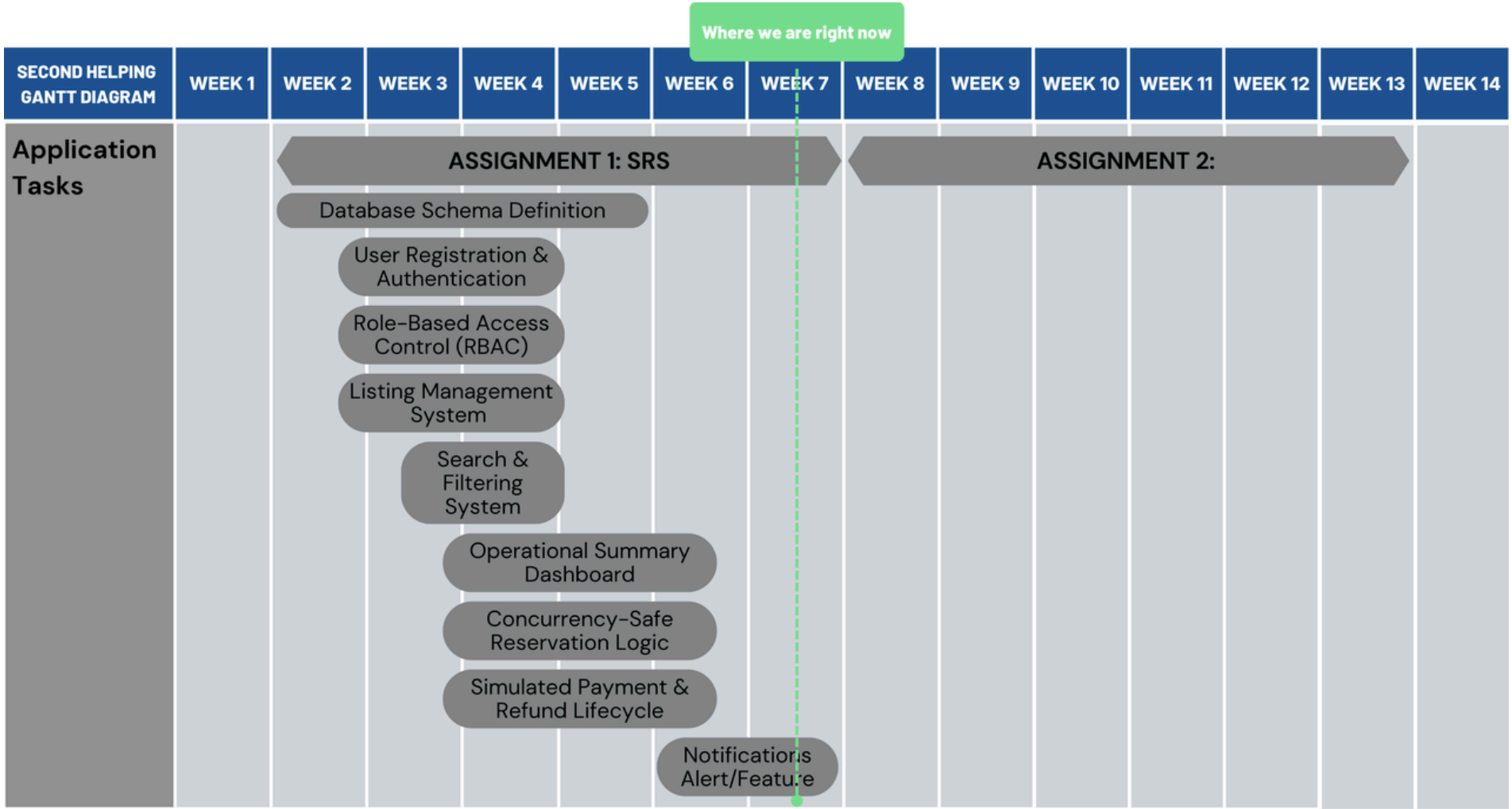
Known limitations section
Simulated payments, in-app-only notifications, local-only reminder endpoint — design decisions vs defects.

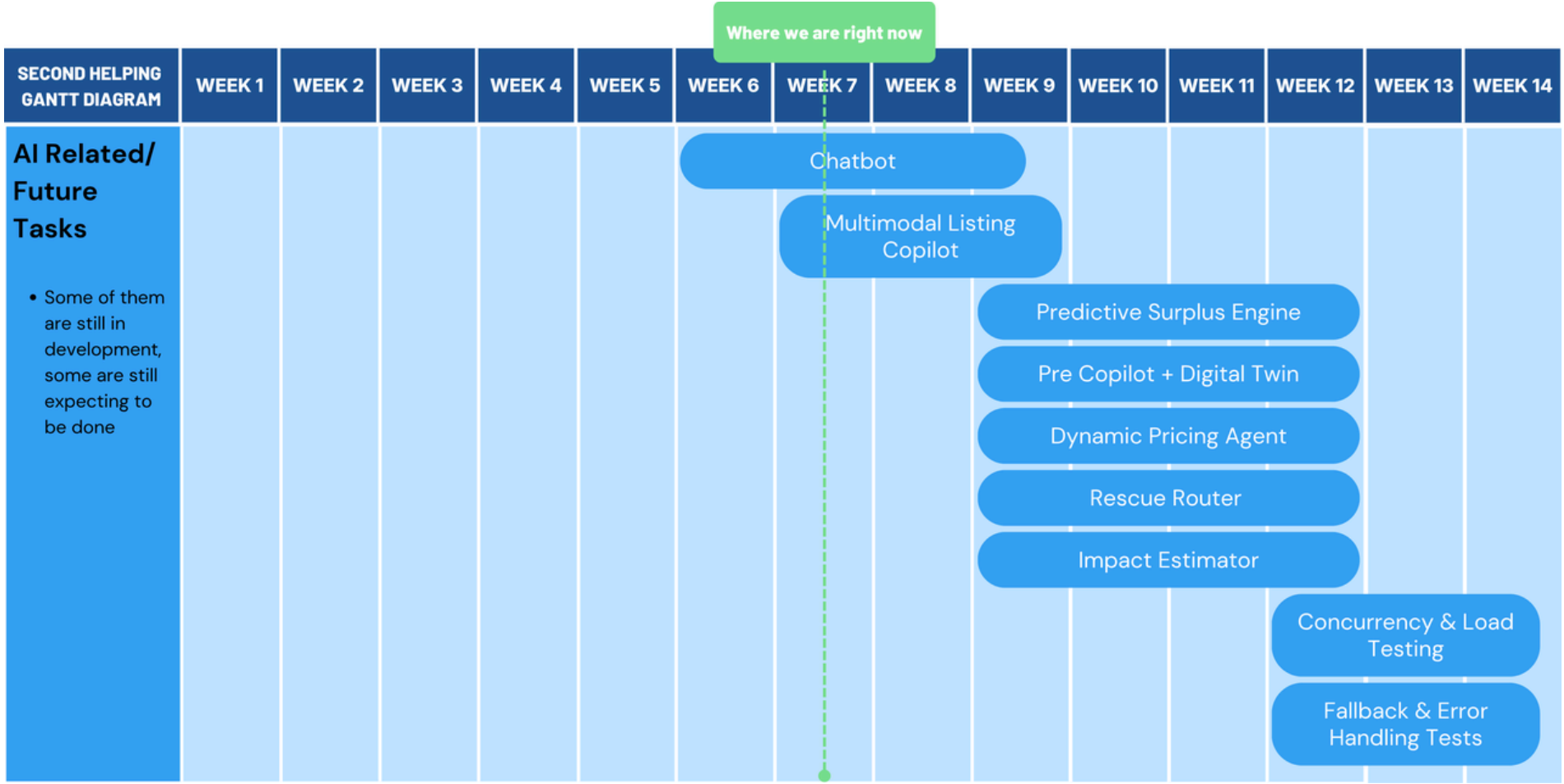
Appendices refreshed
Routes, data model (incl. Notification), dev commands with model pulls.

Source: docs/SRS.md · commits 33cfdf1, a7fa607 · Oct 27


Where we are right now
SECOND HELPING GANTT DIAGRAM
WEEK 1
WEEK 2
WEEK 3
WEEK 4
WEEK 5
WEEK 6
WEEK 7
WEEK 8
WEEK 9
WEEK 10
WEEK 11
WEEK 12
WEEK 13
WEEK 14
Project Tasks
PROJECT TASKS
Project Brainstorming
Role Assignment
AI Modules Role Assignment
Peer-Review Report
Application Tasks
ASSIGNMENT 1: Software Requirements Specification
ASSIGNMENT 2:
Database Schema Definition
User Registration & Authentication
Role-Based Access Control (RBAC)
Listing Management System
Search & Filtering System
Operational Summary Dashboard
Concurrency-Safe Reservation Logic
Simulated Payment & Refund Lifecycle
Notifications Alert/Feature
AI Related/ Future Tasks
• Some of them are still in development, some are still expecting to be done
Chatbot
Multimodal Listing Copilot
Predictive Surplus Engine
Pre Copilot + Digital Twin
Dynamic Pricing Agent
Rescue Router
Impact Estimator
Concurrency & Load Testing
Fallback & Error Handling Tests


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Thank You.

QUESTIONS?

github.com/bhsei/26C

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